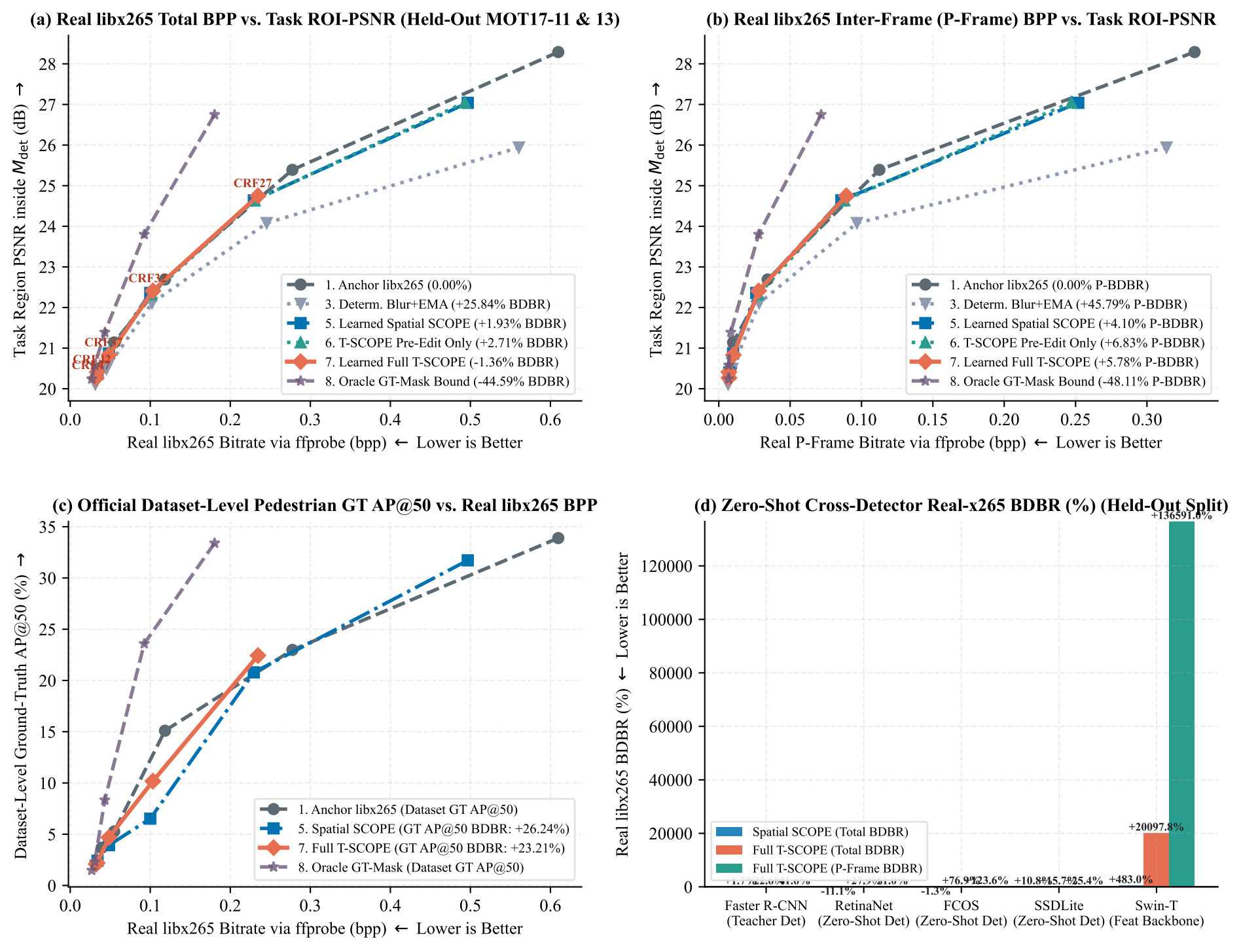

(a) Real libx265 Total BPP vs. Task ROI-PSNR (Held-Out MOT17-11 & 13)
Task Region PSNR inside M_{det} (dB) →
Real libx265 Bitrate via ffprobe (bpp) ← Lower is Better
CRF27
CRF32
CRF37
CRF42
CRF22
1. Anchor libx265 (0.00%)
3. Determ. Blur+EMA (+25.84% BDBR)
5. Learned Spatial SCOPE (+1.93% BDBR)
6. T-SCOPE Pre-Edit Only (+2.71% BDBR)
7. Learned Full T-SCOPE (-1.36% BDBR)
8. Oracle GT-Mask Bound (-44.59% BDBR)
(b) Real libx265 Inter-Frame (P-Frame) BPP vs. Task ROI-PSNR
Task Region PSNR inside M_{det} (dB) →
Real P-Frame Bitrate via ffprobe (bpp) ← Lower is Better
1. Anchor libx265 (0.00% P-BDBR)
3. Determ. Blur+EMA (+45.79% P-BDBR)
5. Learned Spatial SCOPE (+4.10% P-BDBR)
6. T-SCOPE Pre-Edit Only (+6.83% P-BDBR)
7. Learned Full T-SCOPE (+5.78% P-BDBR)
8. Oracle GT-Mask Bound (-48.11% P-BDBR)
(c) Official Dataset-Level Pedestrian GT AP@50 vs. Real libx265 BPP
Dataset-Level Ground-Truth AP@50 (%) →
Real libx265 Bitrate via ffprobe (bpp) ← Lower is Better
1. Anchor libx265 (Dataset GT AP@50)
5. Spatial SCOPE (GT AP@50 BDBR: +26.24%)
7. Full T-SCOPE (GT AP@50 BDBR: +23.21%)
8. Oracle GT-Mask (Dataset GT AP@50)
(d) Zero-Shot Cross-Detector Real-x265 BDBR (%) (Held-Out Split)
Real libx265 BDBR (%) ← Lower is Better
Spatial SCOPE (Total BDBR)
Full T-SCOPE (Total BDBR)
Full T-SCOPE (P-Frame BDBR)
-11.1%
-1.3%
+76.9%
+23.6%
+10.8%
+5.7%
+25.4%
+483.0%
+20097.8%
+136591.6%
Faster R-CNN (Teacher Det)
RetinaNet (Zero-Shot Det)
FCOS (Zero-Shot Det)
SSDLite (Zero-Shot Det)
Swin-T (Feat Backbone)